

16 An object is moving in simple harmonic motion.

Which of the following expression will give a possible variation of its potential energy and kinetic energy with respect to its displacement x , where p , q and r are constants?

	potential energy (PE)	kinetic energy (KE)
A	$PE = px$	$KE = q - rx$
B	$PE = px^2$	$KE = q - rx^2$
C	$PE = \sin px$	$KE = \cos qx$
D	$PE = \sin^2 px$	$KE = \cos^2 qx$